

CHAPTER No 16

PROBLEM No 1

GIVEN:-

$$y = 6 \text{ m} \sin(kx + (600 \text{ rad/s})t + \phi)$$

$$y = 2 \quad y = -2$$

Solution

$$2 = 6 \sin(kx + 600t_1 + \phi) \quad \text{--- (1)}$$

$$-2 = 6 \sin(kx + 600t_2 + \phi) \quad \text{--- (2)}$$

in eq (1)

$$\sin^{-1}(1/3) = kx + 600t_1 + \phi$$

in eq (2)

$$\sin^{-1}(-1/3) = kx + 600t_2 + \phi$$

now sub eq (1) and (2)

$$\sin^{-1}(1/3) - \sin^{-1}(-1/3) = 600(t_1 - t_2)$$

$$t_1 - t_2 = 1 \text{ ns}$$

PROBLEM No 3

$$\lambda = 1.80 \text{ m} \quad \omega = 110 \text{ rad s}^{-1}$$

$$v = ?$$

$$k = ?$$

Solution

$$\text{we know that } k = 2\pi/\lambda \text{ and } v = f\lambda$$

so for k

$$k = 2\pi/1.8 = 3.490$$

$$v = f\lambda \quad \therefore \omega = 2\pi f$$

$$v = 31.54 \quad \therefore f = \omega/2\pi$$

PROBLEM No 10

$$y = 6 \sin(0.02\pi x + 4\pi t)$$

a) $y_m = ?$ b) $\lambda = ?$ c) $f = ?$ d) $v = ?$

$y = ?$ when $x = 3 \text{ s}$ and $t = 0.6$

Solution:-

a) $y_m = 6$

b) we know that

$$k = \frac{2\pi}{\lambda} \text{ so } \lambda = \frac{2\pi}{k} \therefore k = 0.02\pi$$

$$\boxed{\lambda = 1 \text{ m}} \text{ or } \boxed{\lambda = 100 \text{ cm}}$$

c) We know that

$$\omega = 2\pi f \text{ so } f = \frac{\omega}{2\pi} \text{ and } \omega = 4\pi$$

$$f = \frac{4}{2} = 2$$

$$\boxed{f = 2 \text{ Hz}}$$

d) we know that

$$v = \frac{\omega}{k}$$

$$v = \frac{4\pi}{0.02\pi}$$

$$v = 200 \text{ cm/s} = 2 \text{ m/s}$$

e) As the given eq $(0.02\pi x + 4\pi t)$ has a '+' sign for phase then the direction of propagation of wave is along the ^+x axis

f) we know

$$v_{\text{max}} = 2\pi f y_m$$

$$= 2(314)(2)(6)$$

$$\boxed{v_{\text{max}} = 7548}$$

$$y = 6 \sin(0.02\pi(3 \text{ s}) + 4\pi(0.6))$$

$$\boxed{y = -2 \text{ cm}}$$

Problem No 26

$$L = 2.70 \text{ m}$$

$$T = 36 \text{ N}$$

$$m = 260 \text{ g}$$

$$y_m = 7.70 \text{ mm}$$

$$P = 85 \text{ W}$$

Solution

$$P_{\text{avg}} = \frac{1}{2} \mu v \omega^2 y_m^2$$

$$\therefore \mu = \frac{m}{L}$$

$$\mu = 0.096$$

$$P_{\text{avg}} = \frac{1}{2} \mu v 4\pi^2 f^2 y_m^2$$

$$f = \sqrt{\frac{2P}{\mu v 4\pi^2 y_m^2}}$$

$$f = \sqrt{\frac{2P}{\mu \sqrt{\frac{T}{\mu}} 4\pi^2 y_m^2}}$$

$$\therefore v = \sqrt{\frac{T}{\mu}}$$

$$f = 117 \text{ Hz}$$

Problem No 31

$$y'_m = (2y_m) \cos \frac{\phi}{2}$$

$$y'_m = 2y_m \cos \frac{\pi/2}{2}$$

$$= 2y_m \cos \frac{\pi}{4}$$

$$y'_m = \sqrt{2} y_m$$

Problem No 40

$$v = 10 \text{ cm/s}$$

$$\lambda = 1$$

$$\lambda = \frac{v}{f}$$

$$T = 0.5 \text{ s}$$

$$f = \frac{1}{0.5} = 2$$

$$\lambda = \frac{10}{2}$$

we know that

$$v = f \lambda$$

$$\lambda = 5 \text{ cm}$$

PROBLEM No 43

$$L = 10 \text{ m}$$

$$m = 100 \text{ g}$$

$$T = 250$$

Solution

we know that

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$\therefore \lambda = \frac{2L}{n}$$

$$\therefore v = \sqrt{\frac{T}{M}} \quad \therefore M = \frac{v^2}{T}$$

$$\lambda_1 = \frac{2(10)}{1} = 20$$

$$f_1 = \frac{5}{20}$$

$$v = \sqrt{\frac{TL}{m}}$$

$$v = \sqrt{\frac{250 \times 10}{100}}$$

$$\boxed{v = 5 \text{ m/s}}$$